Dual Feature Aggregation Network for Hyperspectral Image Classification

Zhenqiu Shu, Zigao Liu, Zhengtao Yu, and Xiao-Jun Wu

***Abstract*— Recent hyperspectral image (HSI) classification works have focused on developing a promising architecture by combining convolutional neural networks (CNNs) with Transformers. However, most of them fail to consider the interactive fusion of global and local features, thus significantly limiting the HSI classification performance. To address this problem, in this article, we propose a dual feature aggregation network (DFAN) for HSI classification tasks. It effectively aggregates local and global spatial–spectral features to achieve efficient classification. Specifically, we design a dual feature aggregation (DFA) module to extract and aggregate local and global spatial–spectral features. In this module, the lightweight local block is responsible for extracting local spatial–spectral features, and the global block is used to extract global spatial–spectral features and aggregation tasks. Concretely, the token local aggregation multilayer perceptron (TLA-MLP) module in the global block is to extract spatial-position-aware and spectral-channel-aware global discriminative information. Meanwhile, it learns multiscale neighboring token features by aggregating token local neighborhood features. In addition, we employ the local self-global aggregation block to learn the global aggregation features and then fuse them with the local spatial–spectral features. Afterward, the cross-attention aggregation of local and global spatial–spectral features is used to further improve the classification ability of our proposed model. The experimental results on three benchmark datasets show that our proposed DFAN method outperforms other state-of-the-art HSI classification methods.**

***Index Terms*— Cross-attention aggregation, feature aggregation, hyperspectral image (HSI) classification, multiscale neighboring token feature, self-global aggregation, spatial–spectral feature.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of continuous spectral data bands, which can provide rich spatial and spectral features. Until now, HSIs have been widely used in various fields, such as environmental monitoring [1], precision crop farming [2], and geological exploration [3]. In the HSI classification tasks, each pixel of HSIs is treated as a single sample that is categorized based on its feature information. Therefore, how to effectively extract intrinsic spatial–spectral features of HSIs is crucial for classification tasks.

In the early stages, HSI classification methods extract the spectral information hidden in pixels. The most well-known methods include random forest (RF) [4], support vector machine (SVM) [5], and k-nearest neighbor (k-NN) [6]. Since these methods heavily depend on manually designed features, there are significant performance differences across different datasets. Huang et al. [7] integrated dictionary learning with subspace representation into a cohesive model, significantly reducing the complexity of associated optimization problems. Moreover, they fail to achieve promising performance in practical HSI classification applications. With the explosion of deep learning in various fields, many studies have developed extensive deep learning frameworks for HSI classification tasks [8]. In recent years, convolutional neural networks (CNNs) have become one of the most popular frameworks in HSI classification. Hu et al. [9] designed a 1D-CNN HSI classification model to perform classification tasks by directly extracting deep features from pixel spectral vectors. Spectral-based deep learning methods have achieved superior performance compared to most traditional methods. To further consider spatial information of HSIs, several studies utilize a 3-D neighborhood cube of pixels as the input to the network. Zhao and Du [10] designed a 2D-CNN model to extract spatial features from the 3-D patches of the HSIs. Li et al. [11] proposed a 3D-CNN model to extract spatial and spectral features using a single 3-D convolutional kernel. In addition, several HSI classification models have been developed in recent years. Qin et al. [12] introduced a novel domain generalization network that leverages frequency disentanglement and data geometry to maintain semantic consistency and achieve effective domain extension in cross-scene HSI classification. In addition, the approach incorporates adversarial learning and an intermediate domain sampling strategy. Yu et al. [13] attempted to enhance the discriminative ability in hyperspectral image incremental classification (HSIIC) by effectively integrating original and new knowledge. Then, Qin et al. [14] tackled the challenges of biased metaknowledge and domain-specific attributes by employing orthogonal low-rank feature disentanglement and a multiorder spectral interaction block.

To obtain better classification performance by further deepening and widening the networks, Meng et al. [15] proposed a multipath residual network (ResNet) based on ResNet [16]

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for parallel feature extraction. Kang et al. [17] proposed a dual-path network (DPN) to simultaneously construct feature reuse and feed-forward information paths, which takes into account the advantages of ResNet and dense convolutional networks [18]. However, the performances of CNN-based HSI classification methods are usually limited by a single receptive field. To overcome the issue, Lee and Kwon [19] combined multiscale 3-D convolutional kernels to extract multiscale spatial–spectral features of HSIs. Huang et al. [20] designed an adaptive residual convolutional neural network (ARCNN) to merge deep features learned from different receptive fields adaptively. To improve computational efficiency, Cui et al. [21] integrated depthwise convolutions and pointwise convolutions into a 3D-CNN framework to achieve better classification performance with minimal parameter settings. Duan et al. [22] constructed a multiple neighborhood reconstruction (MNR) model and employed hypergraph convolution to integrate local regular and irregular convolutions, thereby enhancing the feature representation for HSI classification. Zhou et al. [23] proposed an attention multihop graph and multiscale convolutional fusion network (AMGCFN) to effectively capture and fuse multilevel information for HSI classification by integrating a multiscale fully CNN and a multihop graph convolutional network (GCN), along with multihead and cross-attention fusion modules. Despite CNN-based models achieving promising performances in HSI classification applications, they are restricted by limited receptive fields, making it challenging to model long-distance dependence of pixels [24].

In recent years, vision Transformer (ViT) [25], which originated from natural language processing (NLP), has made a big splash in the field of computer vision. It also shows strong potential in HSI classification tasks due to its powerful long-distance modeling ability. Although its advantage is to characterize global features, it cannot effectively exploit the local semantics of HSIs [26]. Therefore, various HSI classification models have been proposed by combining the local feature extraction capability of CNNs with the global feature extraction capability of Transformers. Sun et al. [27] proposed a spatial–spectral feature tokenization Transformer (SSFTT) to extract shallow spectral and spatial features using shallow 3D-CNNs and 2D-CNNs. After that, shallow local features are fed into the Transformer module to effectively extract the shallow and deep semantic features of HSIs. Tang et al. [28] used a double-attention structure to fuse local spatial information with global spectral features in each double-attention Transformer encoder module. Chen et al. [29] introduced a fractional Fourier Transform (FrFT) into the network to combine frequency-domain information with spatial–spectral features, thereby amplifying the disparities among various classes. Gao et al. [30] developed a depthwise cross-attention module to extract both self-correlation and cross correlation from multisource feature pairs, thereby highlighting meaningful complementary information and improving classification accuracy. However, the previously mentioned CNN–Transformer methods fail to aggregate local and global information interactively. Therefore, they cannot learn a comprehensive representation of the data, potentially missing essential contextual information crucial for accurate classification. Furthermore, this limitation can hinder the model's capacity to generalize across different scenes or conditions, resulting in reduced classification performance. In addition, these methods ignore the local correlations of domains within the spatial and spectral tokens. Neglecting these domain-specific local correlations can result in the loss of fine-grained details necessary for distinguishing between similar classes.

To deal with this problem, in this article, we propose a novel end-to-end deep classification framework, called dual feature aggregation network (DFAN), for HSI classification tasks. The proposed method consists of a stem module, three dual feature aggregation (DFA) modules, and a head module. Specifically, we design a two-branch stem module, where the 2-D convolutional block generates shallow local features, and the mixed multilayer perceptron (Mix-MLP) is responsible for learning shallow global features. Then, the dual-stream DFA module is utilized to extract the local spatial–spectral and global spatial–spectral information as well as their aggregation and fusion. Finally, local and global spatial–spectral features are fed into the head module to predict the label of each pixel. The experimental results on three benchmark HSI datasets show the effectiveness of the proposed DFAN method in classification tasks.

The main contributions of this work are summarized as follows.

1) In this study, we propose a novel spatial–spectral CNN–Transformer architecture for HSI classification tasks. It extracts interaction local–global spatial–spectral information from HSIs through feature aggregation modeling. Therefore, the spatial and spectral information of HSIs can be fully utilized to further improve HSI classification performance.
2) To integrate local and global information interactively, we introduce a DFA module in our method. Specifically, the local spatial–spectral feature extraction and aggregation interactions are performed by local self-global aggregation, weighted fusion, and cross-attention in DFA. It not only improves the efficiency of feature extraction, but also enhances the model's comprehensive understanding of both local and global spatial–spectral information in HSIs, thereby improving classification performance.
3) To capture the local interdependencies between spatial and spectral domains, the proposed DFA module extracts spatial-position-aware and spectral-channel-aware global features by developing a token local aggregation multilayer perceptron (TLA-MLP) module. In addition, this module performs multiscale neighboring token feature aggregation to improve the generalization capabilities of the model. Extensive experimental results on three benchmark HSI datasets show that our proposed DFAN method achieves better performance than other state-of-the-art HSI classification methods.

The rest of this article is organized as follows. Section II reviews the related work of HSI classification methods. Section III introduces the proposed DFAN approach.

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Section IV presents the experimental results and their analysis. Section V draws this work.

II. Related Work

In this section, we briefly review the HSI classification methods based on CNN and Transformer within the field of HSI classification.

A. CNN-Based HSI Classification Methods

In the past decade, CNNs [31] have experienced rapid growth in HSI classification tasks due to their powerful representation ability. To fully extract spatial and spectral information from HSI, Yang et al. [32] developed a two-branch structure comprising 1D-CNN and 2D-CNN. Residual structures have been commonly used in HSI classification models to mitigate gradient disappearance and network degradation. Paoletti et al. [33] proposed a deep pyramidal ResNet for HSI classification, which forms residual pyramid structures by gradually adjusting convolutional features in the residual blocks. Therefore, it takes into account the spatial and spectral attributes on each pixel of the HSI. To avoid learning features from useless spectral bands and domain pixels, Zhu et al. [34] proposed a residual spectral–spatial attention network, which emphasizes useful bands and neighboring pixels while suppressing the useless ones with spectral attention and spatial attention modules. However, a fixed-size receptive field may limit the capacity of the model to learn multiscale spatial–spectral features, thus hindering its ability to balance fine-grained features with coarse-grained spatial–spectral structural features effectively. Roy et al. [35] introduced an adaptive spatial–spectral kernel attention mechanism in the residual module. It allows the receptive fields to be automatically adjusted with the model updates, thus capturing the cross-channel feature relationships. Gao et al. [36] employed constant mapping to skip some random sublayers, thereby accelerating the training process and preventing overfitting. Although CNN-based HSI classification methods have achieved promising performances on various tasks, they fail to effectively model long-distance dependence within the spatial and spectral features of HSIs.

B. ViT-Based HSI Classification Model

Transformer has recently emerged as a mainstream backbone network that more effectively models the spatial and spectral long-range dependencies of HSIs. However, the original Transformer model cannot fully explore local semantic information among the HSIs. To mitigate this issue, several hybrid CNN–Transformer approaches [37], [38], [39] for the HSI classification applications have also been developed by fully leveraging the advantages of CNN and Transformer, respectively. Qi et al. [40] proposed an integrated global–local 3-D convolutional Transformer (GTCT) network to investigate global–local correlations in both spectral and spatial domains by incorporating 3-D convolution within a dual-branch Transformer framework. The masked autoencoding spectral–spatial Transformer (MAEST) [41] model identifies robust code features by employing mask auto-encoder branches and embedding them in Transformer classification branches, aiming at refining the code features. Wang et al. [42] introduced a novel pipeline to generate high-quality trispectral images from HSIs. To tackle spatial variability issues, it uses pretrained backbones to extract discriminative features. To improve the intraclass consistency of spectral features, Chen et al. [39] constructed a global spectral projection space to generate more robust feature representations, which can enhance the spectral consistency in local neighborhoods. Moreover, a spatial aggregation is further to learn a richer representation of spectral features in the center pixel by fusing all spectral features in the local neighborhoods. Shu et al. [43] proposed a novel dual attention transformer network (DATN) model that embeds local information while extracting global spatial–spectral information by splitting and slicing the original patch smaller. None of them, however, interactively fused global and local spatial–spectral information from HSIs. In addition, they cannot fully explore the local correlation of domains within spatial and spectral tokens.

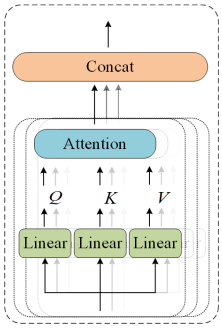


Fig. 1. MHSA.

C. Multihead Self-Attetion

The multihead self-attention (MHSA) [44] mechanism was proposed to capture dependencies between input tokens, especially long-term dependencies that are easily lost in sequence models. As shown in Fig. 1, $\mathbf{X} \in \mathbb{R}^{p\times p\times S}$ is reshaped as tokens $\mathbf{X}' \in \mathbb{R}^{p^2\times S}$, where p^2 is the spatial size of features and S is the channel number of the features. Specifically, $\mathbf{X}'$ is first linearly projected by three randomly initialized learnable weight matrices $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$. Thus, we have

$$\begin{aligned}\mathbf{Q} &= \mathrm{LN}(\mathbf{X}')(\mathbf{W}^Q)^{\mathrm{T}}\\ \mathbf{K} &= \mathrm{LN}(\mathbf{X}')(\mathbf{W}^K)^{\mathrm{T}}\\ \mathbf{V} &= \mathrm{LN}(\mathbf{X}')(\mathbf{W}^V)^{\mathrm{T}}\end{aligned} \tag{1}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V} \in \mathbb{R}^{p^2\times S}$ is the query matrix, key matrix, and value matrix, respectively. LN(.) represents the layer normalization operation that accelerates the convergence rate of the model and enhances its generalization capabilities, simultaneously. Then, the query matrix $\mathbf{Q}$ and the key matrix $\mathbf{K}$ are matched for similarity through the dot product operation. Here, the disappearance of the gradient can be mitigated by dividing the product by $(d_k)^{1/2}$, where $d_k = S$ is the dimension of the pixel tokens. Then, the similarity matrix is obtained by

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the softmax function for normalization. Finally, the similarity matrix and the value matrix $\mathbf{V}$ are multiplied to obtain the result of self-attention. Mathematically, self-attention can be described as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{2}$$

where Softmax(.) represents the softmax function. Furthermore, divide $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ into H heads. Specifically, the output of the ith head is given as follows:

$$\mathbf{h}_i = \text{Softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^{\mathrm{T}}}{\sqrt{d_k}}\right)\mathbf{V}_i. \tag{3}$$

The outputs of all headers are connected as the final output of the MHSA. Thus, we have

$$\text{MHSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{FC}(\text{Concat}(\mathbf{h}_1, \ldots, \mathbf{h}_H)) \tag{4}$$

where Concat(.) and FC(.) denote the concatenation operation and fully connected layer, respectively.

III. Methodology

In this section, we provide a comprehensive description of the proposed DFAN method. Specifically, we first introduce the overall framework of our DFAN model. Subsequently, we describe the DFA module and the TLA-MLP module within our proposed DFAN framework in detail.

A. Overall Framework of DFAN

The overall framework of our DFAN method is illustrated in Fig. 2. It mainly includes three components: the stem, the backbone, and the head. In particular, the first component is responsible for extracting the initial local and global spatial–spectral features from the patches. The backbone component consists of three DFA modules. Its goal is to extract and interactively aggregate discriminative local and global spatial–spectral features. Finally, the global and local features are fed into a head module composed of a global average pooling (GAP) and a linear projection operation to recognize the land cover categories.

Here, the available pixels of HSIs and their labels are divided into a training set and a testing set. Suppose that the HSI dataset consists of N available image pixels $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$, $\mathbf{x}_i \in \mathbb{R}^B$ and their labels $\{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N\}$, $\mathbf{y}_i \in \mathbb{R}^C$, where B represents the numbers of spectral bands and C denotes the number of the land cover categories. To fully utilize the spectral and spatial contexts hidden in the HSIs, neighboring pixels of each pixel are used to form a 3-D patch $\mathbf{P} \in \mathbb{R}^{w\times w\times B}$ and then feed it into the classification model, where $w \times w$ represents the spatial size of patches.

To avoid loss and distortion of the original spatial–spectral feature information, we directly input patch $\mathbf{P} \in \mathbb{R}^{w\times w\times B}$ into the network. First, the local and global spatial–spectral features are extracted in the stem module. These processes can be formulated as follows:

$$\begin{aligned}\mathbf{X}_{\text{local}} &= \text{Conv}_{3\times 3}(\mathbf{P})\\ \mathbf{X}_{\text{global}} &= \text{MM}(\mathbf{P})\end{aligned} \tag{5}$$

where $\mathbf{X}_{\text{local}}$ and $\mathbf{X}_{\text{global}}$ refer to the local and global spatial–spectral features, respectively. $\text{Conv}_{3\times 3}$ presents the 3×3 convolution operation. MM(.) denotes the Mix-MLP, which can be formulated as follows:

$$\begin{aligned}\text{MM}(\mathbf{P}) = \text{Trbac}_{\text{spa}}\big(\text{FC}_1\big(\text{Trans}_{\text{spa}}\big(\text{Trbac}_{\text{spc}}(\text{FC}_2\\ \times\big(\text{Trans}_{\text{spc}}(\mathbf{P})\big)\big)\big)\big)\big)\end{aligned} \tag{6}$$

where $\text{Trans}_{\text{spc}}(.)$ and $\text{Trans}_{\text{spa}}(.)$ represent the feature compression and transpose operations that convert input feature $\mathbf{Z} \in \mathbb{R}^{p\times p\times S}$ into $\mathbf{Z}_{\text{spc}} \in \mathbb{R}^{p^2\times S}$ and $\mathbf{Z}_{\text{spa}} \in \mathbb{R}^{S\times p^2}$. Correspondingly, $\text{Trbac}_{\text{spc}}(.)$ and $\text{Trbac}_{\text{spa}}(.)$ are the opposite operation to $\text{Trans}_{\text{spc}}(.)$ and $\text{Trans}_{\text{spa}}(.)$. After that, the two initial features are input into the DFA modules for deep semantic feature extraction. Finally, the discriminative global and local spectral–spatial features are concatenated, and subsequently fed into the classifier after a global pooling operation to obtain the final categories.

B. DFA Module

To improve feature extraction efficiency and achieve a comprehensive understanding of local and global spatial–spectral information in HSIs, we incorporate a DFA module inspired by Chu et al. [45] into our framework. It consists of a lightweight local branch and a global aggregation branch. They are employed to refine the spatial–spectral semantic features and perform several aggregations and fusions. Similar to ConvNeXt [46], we design a local block to extract local features, which can significantly reduce computational costs. Mathematically, it can be represented as follows:

$$\mathbf{X}'_{\text{local}} = \mathbf{X}_{\text{local}} + \text{PW}_2(\text{PW}_1(\text{DW}(\mathbf{X}_{\text{local}}))) \tag{7}$$

where DW(.) is the depthwise convolution operation. $\text{PW}_1(.)$ and $\text{PW}_2(.)$ are the pointwise convolution operation. It is worth noting that we employ a batch normalization operation and a nonlinear activation function after each convolution. And $\mathbf{X}'_{\text{local}}$ containing local information is the output of the lightweight convolution block. Meanwhile, the TLA-MLP module transforms the input global features into global position-channel aware tokens, which can be expressed as follows:

$$\mathbf{X}^{\text{ga}}_{\text{global}} = \text{PW}_3\big(\text{TLA-MLP}\big(\mathbf{X}_{\text{global}}\big)\big). \tag{8}$$

Afterward, the global position-channel aware tokens are passed to the aggregation and fusion part. First, we adopt a local self-global aggregation strategy to aggregate the local spatial–spectral information extracted by a lightweight local block. Specifically, the global aggregation features $\mathbf{X}^{\text{ga}}_{\text{local}}$ are formed by applying a local self-global aggregation block to local features $\mathbf{X}'_{\text{local}}$. The calculations for this block are given as follows:

$$\begin{aligned}\mathbf{Q}_{\text{ga}} &= \text{LN}\big(\mathbf{X}'_{\text{local}}\big)\big(W^{Q_{\text{local}}}\big)^{\mathrm{T}}\\ \mathbf{K}_{\text{ga}} &= \text{LN}\big(\mathbf{X}'_{\text{local}}\big)\big(W^{K_{\text{local}}}\big)^{\mathrm{T}}\\ \mathbf{V}_{\text{ga}} &= \text{LN}\big(\mathbf{X}'_{\text{local}}\big)\big(W^{V_{\text{local}}}\big)^{\mathrm{T}}\\ \mathbf{X}^{\text{ga}}_{\text{local}} &= \text{MHSA}\big(\mathbf{Q}_{\text{ga}}, \mathbf{K}_{\text{ga}}, \mathbf{V}_{\text{ga}}\big).\end{aligned} \tag{9}$$

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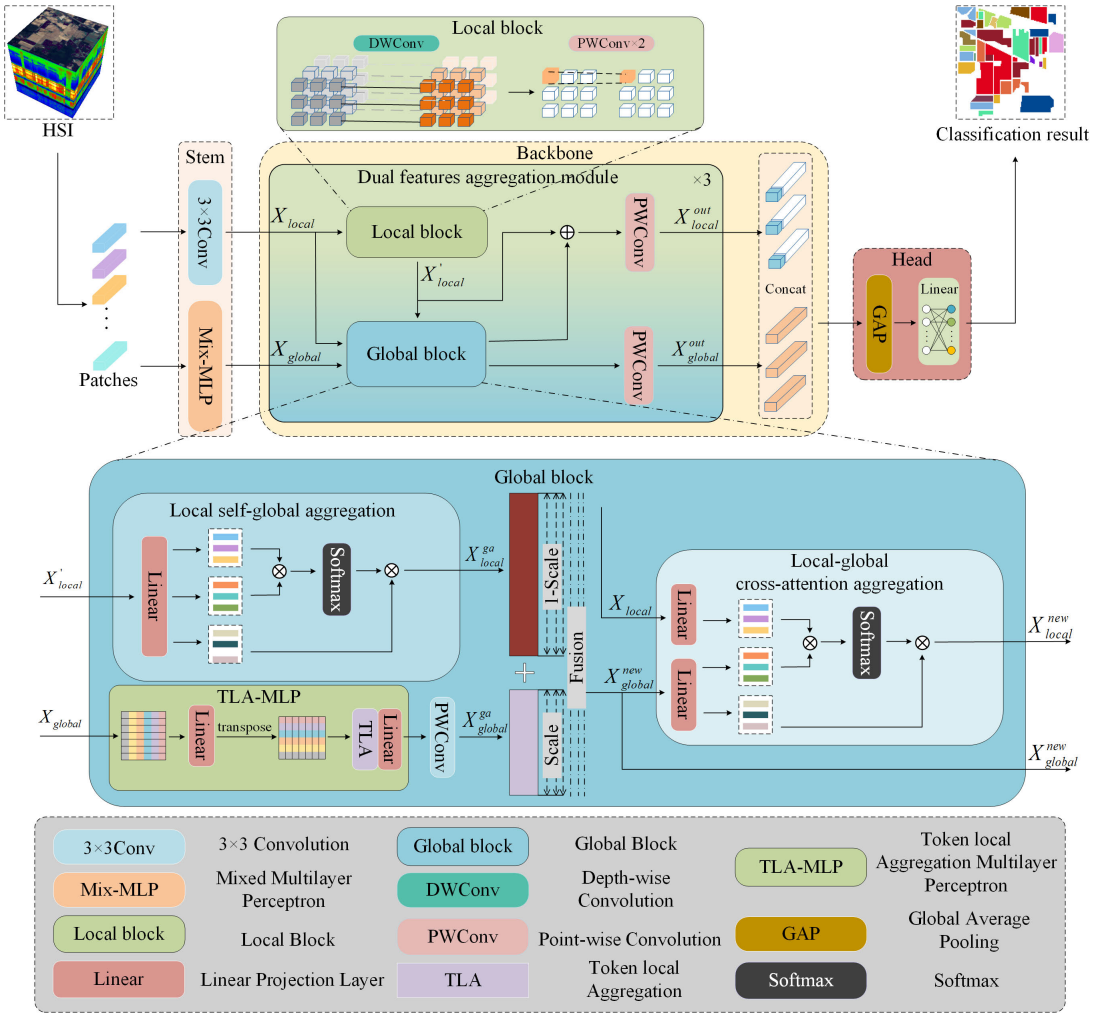


Fig. 2. Overall framework of the proposed DFAN method.

To fully exploit the global information and enhance the information interaction between global and local branches, we adopt a weighted sum approach to fuse the global position-channel aware tokens $\mathbf{X}_{\text{global}}^{\text{ga}}$ and the global aggregation features $\mathbf{X}_{\text{local}}^{\text{ga}}$ as follows:

$$\mathbf{X}_{\text{global}}^{\text{new}} = \text{Fuse}\left(\mathbf{X}_{\text{global}}^{\text{ga}}, \mathbf{X}_{\text{local}}^{\text{ga}}\right) = \alpha \mathbf{X}_{\text{global}}^{\text{ga}} + (1-\alpha)\mathbf{X}_{\text{local}}^{\text{ga}} \tag{10}$$

where $\mathbf{X}_{\text{global}}^{\text{new}}$ refers to the fused global features, and the hyperparameter $\alpha \in [0, 1]$ represents the fusion scale. To further enrich the local features, the DFA module inputs local features $\mathbf{X}_{\text{global}}$ and $\mathbf{X}_{\text{global}}^{\text{new}}$ that interact aggregation through the local–global cross-attention aggregation block, which can be mathematically given as follows:

$$\begin{aligned}
\mathbf{Q}_l &= \text{LN}(\mathbf{X}_{\text{local}})\left(\mathbf{W}^{X'}\right)^{\text{T}} \\
\mathbf{K}_g &= \text{LN}\left(\mathbf{X}_{\text{global}}^{\text{new}}\right)\left(\mathbf{W}^{K'}\right)^{\text{T}} \\
\mathbf{V}_g &= \text{LN}\left(\mathbf{X}_{\text{global}}^{\text{new}}\right)\left(\mathbf{W}^{V'}\right)^{\text{T}} \\
\mathbf{X}_{\text{local}}^{\text{new}} &= \text{MHSA}\left(\mathbf{Q}_l, \mathbf{K}_g, \mathbf{V}_g\right).
\end{aligned} \tag{11}$$

By adopting the pointwise convolution operation, the outputs of two branches in the DFA module are obtained as follows:

$$\begin{aligned}
\mathbf{X}_{\text{local}}^{\text{out}} &= \text{PW}_4\left(\mathbf{X}'_{\text{local}} + \mathbf{X}_{\text{local}}^{\text{new}}\right) \\
\mathbf{X}_{\text{global}}^{\text{out}} &= \text{PW}_5\left(\mathbf{X}_{\text{global}}^{\text{new}}\right).
\end{aligned} \tag{12}$$

C. Token Local Aggregation MLP

To enhance the model's generalization by aggregating multiscale neighboring token features, we propose the TLA-MLP module to effectively extract spatial-position-aware and spectral-channel-aware global features. Within the TLA-MLP module, inspired by Guo et al. [47], the extraction of global spatial–spectral features is complemented by the TLA module, which allows tokens to aggregate the spatial–spectral semantic features of their local spatial neighbors. Fig. 3 shows the structure of the TLA-MLP module. Mathematically, it can be expressed as follows:

$$\begin{aligned}
\mathbf{X}_{\text{TLA-MLP}} = \text{Trbac}_{\text{spa}}\big(\text{FC}_3\big(\text{Trans}_{\text{spa}}(\text{TLA} \\
\times\big(\text{Trbac}_{\text{spc}}\big(\text{FC}_4\big(\text{Trans}_{\text{spc}}\big(\mathbf{X}_{\text{global}}\big)\big)\big)\big)\big)\big)
\end{aligned} \tag{13}$$

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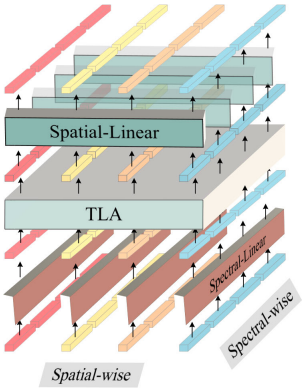


Fig. 3. Structure of the proposed TLA-MLP module.

where TLA(.) denotes the TLA module. The TLA-MLP model consists mainly of two linear layers, two transpose operations and a TLA block. Due to the characteristics of the linear layer, TLA-MLP can effectively capture spatial-position and spectral-channel information. The linear layer and the transpose operation work together to extract global spatial–spectral features, and the TLA block is used to perform multiscale feature aggregation.

The spatial features with different dilation rates are usually different from one another. Therefore, we seek to adaptively aggregate multiscale pooling features with different dilation rates. Specifically, as shown in Fig. 4, we use spatial average pooling with different dilation rates to aggregate multiscale local spatial neighborhood features from different pooled fields.

Here, the TLA modules generally design K branches. $d = 0$ denotes the identity branch, where neither pooling is used. $d = 1$ denotes a 3×3 average pooling branch, and $d = K - 1$ denotes the average pooling branches with the dilation rate $K - 2$. The multibranch outputs $\mathbf{T} \in \mathbb{R}^{p\times p\times C\times K}$ in TLA are concatenated as follows:

$$\mathbf{T} = \text{Concat}\left(\mathbf{T}^0, \ldots \mathbf{T}^{K-1}\right) \tag{14}$$

where Concat(.) denotes the concatenation operation. Specifically, a dilation feature selector in the TLA module aims at aggregating multiscale pooling features with different dilation rates to generate a dilation attention mask. Thus, we have

$$\text{Mask} = \text{Softmax}\left(\text{Conv}_{3\times 3}\left(\text{Conv}_{3\times 3}\left(\mathbf{X}_{\text{global}}\right)\right)\right) \tag{15}$$

where Mask $\in \mathbb{R}^{p\times p\times K}$ denotes the spatial dilation attention mask, and Softmax(.) represents the Softmax function. Subsequently, the broadcast mechanism is utilized to multiply different dilation rate features and masks by elements as follows:

$$\mathbf{T}_c^{\text{out}'} = \text{Mask} \otimes (\mathbf{T}_c) \tag{16}$$

where $\mathbf{T}^{\text{out}'}$ represents the multiscale aggregated features, and $\mathbf{T}_c \in \mathbb{R}^{p\times p\times K}$ and $\mathbf{T}_c^{\text{out}'} \in \mathbb{R}^{p\times p\times K}$ represent the cth channel of $\mathbf{T}$ and $\mathbf{T}^{\text{out}'}$, respectively. Finally, $\mathbf{T}^{\text{out}'} \in \mathbb{R}^{p\times p\times C\times K}$ summing the features in the last dimension yields the output features of the TLA, $\mathbf{T}^{\text{out}} \in \mathbb{R}^{p\times p\times C}$.

IV. Experiment and Analysis

In this section, we verify the effectiveness of the proposed DFAN method by conducting some experiments on three benchmark HSI datasets. Thus, we compare several state-of-the-art HSI classification approaches based on different metrics.

A. HSI Datasets

The Loukia dataset is a subdataset of the HyRANK hyperspectral dataset, which was developed in the framework of the ISPRS Scientific Initiatives [48]. The HyRANK dataset contains HSI data from satellites acquired using the Hyperion sensor (EO-1 and USGS). The Loukia dataset includes 176 spectral channels, with a spatial resolution of 249 $\times$ 945 pixels. In addition, the Loukia dataset contains 14 land-cover classes.

The WHU-Hi-LongKou dataset was collected between 13:49 and 14:37 on July 17, 2018, in Longkou Town, Hubei Province, China, using an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) unmanned aerial vehicle (UAV) platform. During the data collection, the weather was clear and cloudless, with a temperature of approximately 36 °C, and a relative air humidity of about 65%. The study area was a simple agricultural landscape featuring six crop species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. The UAV flew at an altitude of 500 m, capturing imagery of size 550 $\times$ 400 pixels with 270 spectral bands ranging from 400 to 1000 nm. The spatial resolution of these HSIs is approximately 0.463 m.

The Houston 2013 (HUST 2013) dataset was collected by the ITRES CASI-1500 sensor in June 2012, which was provided by IEEE GRSS Data Fusion Competition 2013 [49]. This dataset contains 144 spectral bands with a spatial resolution of 349 $\times$ 1905 pixels and includes 15 land-cover classes.

The proposed method is fairly evaluated by removing all unlabeled samples and randomly selecting 3%, 3%, and 94% of the whole labeled samples as training, validation, and testing sets, respectively, for the Loukia and HUST 2013 datasets while 0.1%, 0.1%, and 99.8% for the WHU-Hi-Longkou dataset.

The experiments are conducted with the following hardware configuration: Intel Xeon Silver 4214R CPU, Nvidia GeForce RTX 4090 GPU, and 128 GB of RAM. The software setup includes Windows 10, Python 3.8, PyTorch 2.1.1, Scikit-learn 1.3.2, and Cuda 12.2.

We set the patch size and the batch size to 9 $\times$ 9 and 64, respectively, for all HSI datasets. In addition, the maximum training epoch for the comparison methods is set to 100. The momentum optimizer and the learning rate decay strategy are used to accelerate the convergence of the model. Specifically, the momentum and weight decay values are empirically set to 0.8 and 0.0001, respectively.

To quantitatively verify the effectiveness of the proposed method, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are used as metrics. Generally,

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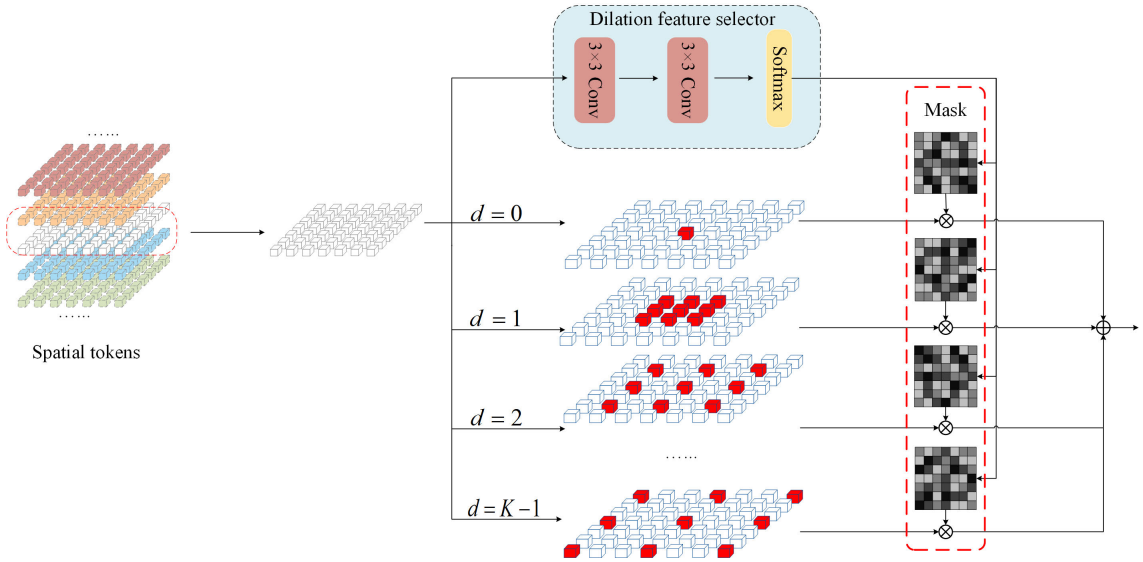


Fig. 4. Structure of the proposed TLA module.

TABLE I
RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE LOUKIA DATASET

No.	DFAN	3D-CNN	SSAN	MLNet-A	ViT	morphFormer	MAEST	MASSFormer	SSFTT
1	74.74±6.71	58.67±6.40	41.86±4.00	71.10±3.37	43.71±17.10	52.94±4.24	71.10±3.37	55.11±16.66	61.49±5.82
2	99.68±0.71	83.38±10.66	90.78±8.63	99.05±1.42	0.00±0.00	100.00±0.00	99.05±1.42	96.72±3.82	91.93±7.94
3	82.26±3.63	71.76±3.34	70.15±3.82	88.96±2.70	66.35±1.67	78.77±4.41	88.96±2.70	75.57±5.49	71.49±2.27
4	71.43±26.24	75.92±23.59	22.50±8.92	20.00±44.72	0.00±0.00	37.00±34.93	20.00±44.72	10.33±14.28	38.85±13.03
5	82.84±2.65	78.63±2.04	74.51±3.19	81.85±0.93	65.49±1.06	79.74±1.46	81.85±0.93	75.89±9.06	78.64±1.23
6	70.18±12.05	62.51±5.45	54.00±12.27	65.18±6.34	86.70±4.48	61.77±14.15	65.18±6.34	52.77±14.23	70.36±7.29
7	74.20±4.33	77.73±4.87	66.76±5.58	75.48±2.53	76.16±2.13	63.32±5.80	75.48±2.53	64.09±7.44	71.50±2.93
8	74.94±1.74	70.79±2.11	71.32±2.97	73.14±1.13	60.72±0.33	72.30±2.99	73.14±1.13	62.98±3.45	72.06±0.52
9	79.12±0.64	72.65±1.31	69.20±2.53	75.09±0.85	64.40±0.75	76.51±1.65	75.09±0.85	71.20±3.19	78.78±0.97
10	81.54±1.28	74.77±1.24	73.77±2.58	76.63±0.76	67.21±1.79	75.38±1.88	76.63±0.76	70.52±2.36	79.43±1.27
11	91.08±3.18	67.22±6.57	74.36±6.91	94.71±0.71	60.60±7.26	85.58±2.77	94.71±0.71	54.29±13.58	79.48±3.90
12	93.33±3.46	92.57±1.79	86.33±5.06	96.32±2.04	72.48±0.36	92.56±4.35	96.32±2.04	90.83±4.38	92.81±3.43
13	100.00±0.00	99.92±0.09	99.94±0.10	100.00±0.00	97.64±0.49	98.60±1.43	100.00±0.00	99.95±0.04	99.85±0.30
14	99.76±0.29	99.86±0.13	97.51±2.61	99.86±0.21	99.43±0.13	99.81±0.42	99.86±0.21	99.03±1.68	99.34±0.45
OA(%)	**83.17±0.53**	77.87±0.70	74.80±1.09	81.13±0.33	70.13±0.56	79.30±0.62	81.13±0.33	74.93±2.35	80.71±0.25
AA(%)	**83.94±1.63**	77.60±2.26	70.93±2.24	79.81±3.34	61.49±1.53	76.73±3.01	79.81±3.34	69.95±3.13	77.57±1.70
Kappa(%)	**79.89±0.61**	73.41±0.86	69.89±1.41	77.36±0.41	63.56±0.78	75.28±0.76	77.36±0.41	69.94±2.88	77.01±0.32
FLOPs(G)	6.55	0.30	5.86	14.46	67.16	2.74	14.46	10.54	2.85
Params(M)	0.63	0.20	1.82	1.39	6.40	0.18	1.39	0.98	0.82
TrainTime(s)	52.68	12.18	24.06	14.68	21.56	32.94	22.64	19.99	20.04
TestTime(s)	4.26	66.01	0.63	1.19	1.60	2.17	1.35	1.33	1.06

the higher these values, the better the classification performance. Additionally, we introduce floating point operations per second (FLOPs) and the number of model parameters (Params) to analyze the complexity of the proposed method and its competitors. The proposed models are trained five times during the training process, and then, the parameters with the highest OA are selected as the final parameters of the training model. This final model is used to predict the category of each pixel in the HSIs and then produce a feature map.

B. Classification Performance

To verify the superiority of the proposed model, we employ several state-of-the-art HSI classification models, such as 3D-CNN [11], SSAN [50], MLNet-A [51], ViT [25], morphFormer [52], MAEST [41], MASSFormer [53], and SSFTT [27], as the comparison methods. The first three HSI classification methods are based on CNNs, while the remaining methods belong to the Transformer structure.

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TABLE II
RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE HUST 2013 DATASET

No.	DFAN	3D-CNN	SSAN	MLNet-A	ViT	morphFormer	MAEST	MASSFormer	SSFTT
1	94.12±0.55	97.26±1.18	94.65±2.92	93.27±0.36	94.44±3.13	93.93±0.63	93.27±0.36	97.46±2.93	95.93±1.45
2	98.91±1.05	88.15±4.19	91.77±4.60	95.42±2.44	74.92±2.95	92.87±3.65	95.42±2.45	82.24±4.13	91.96±4.18
3	100.00±0.00	96.23±5.88	96.98±2.73	100.00±0.00	62.02±2.04	99.40±1.09	100.00±0.00	97.94±1.44	98.59±1.11
4	99.53±0.44	97.34±0.72	97.39±0.80	99.80±0.10	79.86±0.90	99.37±0.70	99.80±0.10	97.35±2.38	97.67±1.51
5	96.34±1.48	91.99±2.51	93.43±1.78	96.45±1.81	89.01±3.14	95.04±0.90	96.45±1.81	94.50±0.85	95.58±0.58
6	99.84±0.21	95.30±2.74	94.00±3.93	100.00±0.00	97.10±3.03	99.04±0.83	100.00±0.00	92.76±4.25	88.39±3.80
7	92.82±2.68	83.44±3.37	88.09±1.81	88.41±1.38	80.81±0.91	88.52±3.73	88.41±1.38	81.88±6.71	85.51±2.28
8	97.86±0.86	78.53±6.52	94.06±2.16	97.01±0.88	69.50±2.97	96.63±1.21	97.01±0.88	93.46±2.00	92.90±2.22
9	93.70±0.74	78.53±6.52	90.31±1.55	90.46±1.96	78.36±2.43	88.94±2.07	90.46±1.96	87.28±3.33	88.52±3.62
10	92.99±1.56	60.74±3.99	88.38±5.12	83.22±2.23	48.62±4.84	88.51±4.28	83.22±2.23	87.53±4.84	90.54±1.05
11	93.44±1.94	67.87±8.10	81.38±4.52	87.58±1.34	47.91±2.49	89.38±2.22	87.58±1.14	80.98±13.36	91.41±1.39
12	89.62±0.98	67.09±4.73	84.17±2.80	89.61±1.48	60.88±4.73	91.03±1.88	89.61±1.48	86.71±7.41	88.92±1.26
13	93.30±1.63	83.42±8.75	85.48±2.53	96.03±1.58	66.22±13.65	89.58±4.83	96.03±1.58	87.36±5.02	89.36±1.89
14	97.50±1.82	96.49±5.33	92.93±2.78	99.60±0.41	69.21±1.51	96.11±4.06	99.60±0.41	97.68±2.23	96.83±2.01
15	99.61±0.27	97.77±2.87	97.04±1.91	99.65±0.13	91.92±5.15	99.11±0.91	99.65±0.13	98.41±0.70	99.01±0.64
OA(%)	**95.39±0.32**	83.48±1.63	90.76±1.21	93.15±0.72	72.24±0.75	93.05±0.76	93.15±0.72	89.46±2.56	92.43±0.44
AA(%)	**95.97±0.29**	86.03±1.72	91.34±1.04	94.43±0.61	74.05±1.07	93.83±0.83	94.43±0.61	90.90±1.57	92.74±0.42
Kappa(%)	**95.02±0.35**	82.14±1.77	90.01±1.31	92.59±0.78	69.97±0.82	92.48±0.82	92.59±0.78	88.60±2.77	91.81±0.48
FLOPs(G)	6.24	0.24	5.07	14.08	66.99	2.25	14.08	8.94	2.33
Params(M)	0.60	0.16	1.75	1.36	6.39	0.15	1.36	0.83	0.67
TrainTime(s)	41.02	7.53	26.79	18.55	22.20	31.49	23.06	19.07	19.88
TestTime(s)	3.03	0.69	1.47	1.31	2.09	2.47	1.30	1.10	1.08

TABLE III
RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE WHU-HI-LONGKOU DATASET

No.	DFAN	3D-CNN	SSAN	MLNet-A	ViT	morphFormer	MAEST	MASSFormer	SSFTT
1	97.92±1.30	93.62±3.67	94.63±0.85	97.15±1.54	95.97±0.81	95.02±1.73	97.15±1.54	88.82±6.13	93.03±1.51
2	90.35±6.69	50.54±5.88	77.33±1.44	74.91±1.19	54.92±0.59	64.71±13.61	74.91±1.19	54.70±12.44	73.98±5.62
3	98.54±2.71	32.37±42.26	90.30±11.25	100.00±0.00	0.00±0.00	40.00±54.77	100.00±0.00	20.00±44.72	64.80±16.95
4	95.10±1.39	86.31±0.74	94.27±1.25	92.33±1.04	88.08±0.16	89.77±2.03	92.33±1.03	90.07±3.17	93.38±1.44
5	76.06±12.40	32.81±11.60	66.17±5.61	85.49±4.62	42.25±14.09	43.47±29.21	85.49±4.62	24.89±34.08	71.93±8.06
6	99.60±0.62	95.86±3.43	92.11±7.19	99.92±0.08	99.42±0.34	97.73±3.68	99.92±0.08	97.48±3.13	97.58±2.88
7	99.75±0.13	97.19±1.68	99.39±0.37	98.94±0.13	98.35±0.05	99.29±0.35	98.94±0.13	98.63±0.22	98.36±0.62
8	89.79±4.41	74.94±12.07	73.79±1.49	83.91±1.37	60.60±0.75	71.01±13.89	83.91±1.37	69.73±8.65	84.34±5.66
9	93.18±2.26	72.29±18.36	63.65±9.14	94.21±2.89	58.06±2.87	75.33±6.03	94.21±2.89	89.57±5.68	83.18±3.91
OA(%)	**96.47±0.57**	88.95±0.96	93.24±0.87	94.67±0.27	89.94±0.12	91.82±1.45	94.67±0.27	89.97±3.21	93.12±0.76
AA(%)	**93.36±1.07**	70.66±6.03	83.51±2.64	91.87±0.57	66.42±1.54	75.15±10.37	91.87±0.57	70.43±11.53	84.51±2.76
Kappa(%)	**95.33±0.76**	85.23±1.28	91.08±1.13	92.91±0.36	86.60±0.15	89.11±1.96	92.91±0.36	86.65±4.28	90.90±1.01
FLOPs(G)	7.18	0.47	8.20	15.58	67.66	4.18	15.58	15.24	4.38
Params(M)	0.69	0.30	2.01	1.50	6.45	0.25	1.50	1.41	1.25
TrainTime(s)	28.10	5.22	13.44	11.00	11.34	17.94	12.42	25.82	12.82
TestTime(s)	73.91	13.39	26.99	20.81	20.24	40.50	21.80	88.18	19.27

For a fair comparison, all HSI classification methods are run on the same training and testing sets. Tables I and II record the classification results of our DFAN model and other comparison models on the Loukia, HUST 2013, and WHU-Hi-Longkou datasets. Figs. 5–7 show the classification maps of different models on three datasets. From the experimental results on three benchmark HSI datasets, we can draw the following observations.

1) It can be seen that the proposed DFAN model achieves the highest classification performance on all datasets. As shown in Table I, our proposed DFAN method achieves the highest average OA value (83.17%), AA value (83.94%), and Kappa (79.89%) among all HSI classification models. The average OA value of the proposed DFAN model improves by 5.30% (3D-CNN), 8.37% (SSAN), 2.04% (MLNet-A), 13.04%

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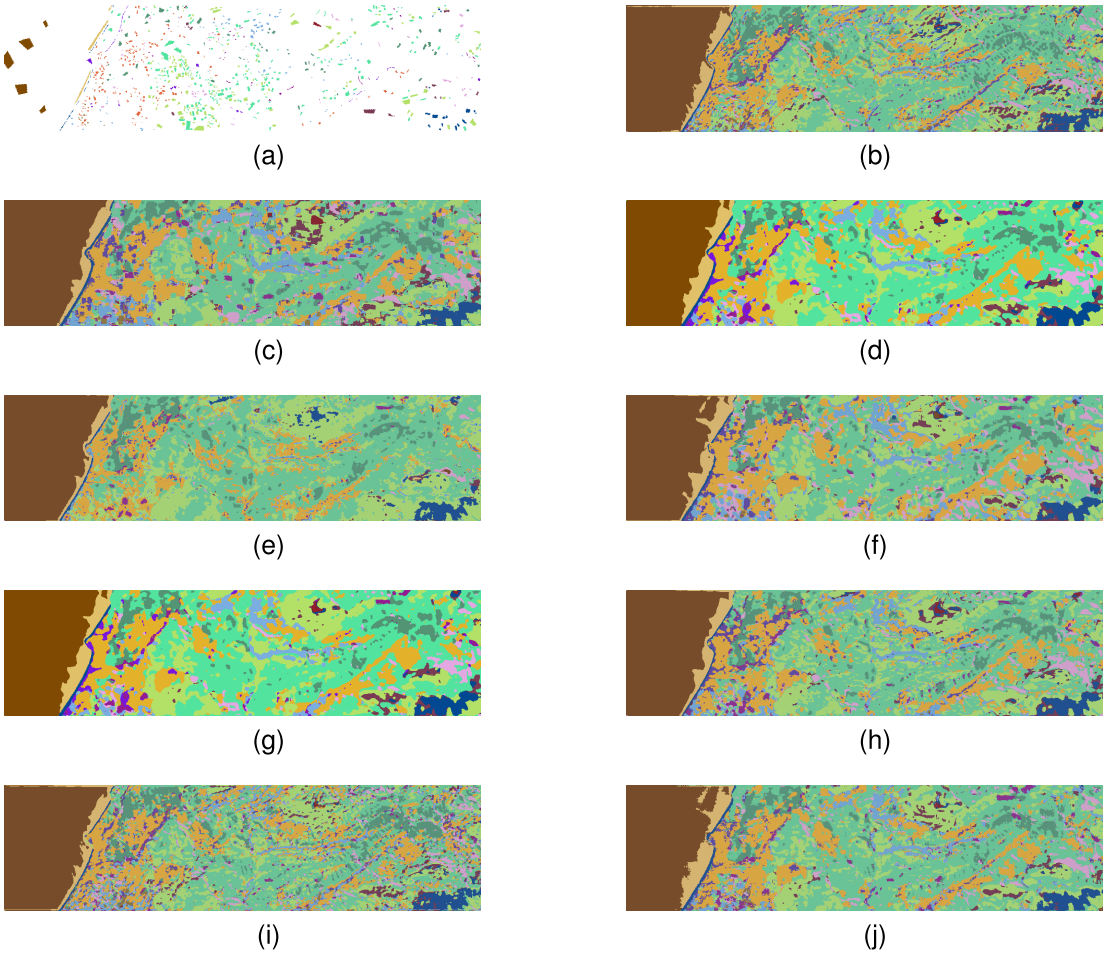


Fig. 5. Classification maps of different HSI classification models on the Loukia dataset. (a) Ground truth. (b) 3D-CNN. (c) SSAN. (d) MLNet-A. (e) ViT. (f) morphFormer. (g) MAEST. (h) MASSFormer. (i) SSFTT. (j) DFAN.

(ViT), 3.87% (morphFormer), 2.04% (MAEST), 8.24% (MASSFormer), and 2.46% (SSFTT), respectively. Table II displays the performances of different models on the HUST 2013 dataset. It is clear to see that our DFAN model achieves the highest average OA value (95.39%), AA value (95.97%), and Kappa value (95.02%). The improvement of OA values obtained by the proposed DFAN model is 11.91% (3D-CNN), 4.63% (SSAN), 2.24% (MLNet-A), 23.15% (ViT), 2.34% (morphFormer), 2.24% (MAEST), 5.93% (MASSFormer), and 2.96% (SSFTT), respectively. Table III shows the classification results of different methods on the WHU-Hi-Longkou dataset. Our DFAN model also achieves the highest OA value (96.47%), AA value (93.36%), and Kappa value (95.33%) on the WHU-Hi-Longkou dataset. In terms of OA metric, our proposed DFAN model has increased by 7.52% (3D-CNN), 3.23% (SSAN), 1.80%(MLNet-A), 6.53% (ViT), 4.65% (morphFormer), 1.80% (MAEST), 6.50% (MASSFormer), and 3.35% (SSFTT), respectively. It can be seen that both 3D-CNN and ViT obtain lower classification performances than other methods with limited training samples. In addition, our DFAN method achieves the best classification performance among all approaches on different HSI datasets. The main reason is that it can effectively extract local–global spatial–spectral information from HSIs through feature aggregation modeling.

2) Among all HSI classification models, the classification maps of both 3D-CNN and ViT exhibit the most misclassified pixels and blurred boundaries between classes. Our DFAN method produces the classification maps that are the most accurate results on all datasets, which is consistent with the results shown in Tables I–III. Specifically, compared to other HSI classification models, the proposed DFAN method exhibits the fewest misclassification points and the smoothest boundaries between different categories. The results further demonstrate the superiority of our DFAN method in HSI classification tasks.
3) It can be seen that the proposed DFAN method exhibits moderate levels of FLOPs and parameters among all HSI classification methods. However, our DFAN method requires more training and testing times than other competitors in most cases. The primary reason is that the global block involves many time-consuming matrix multiplication and softmax operations. Although the DFAN model cannot achieve optimal training and testing

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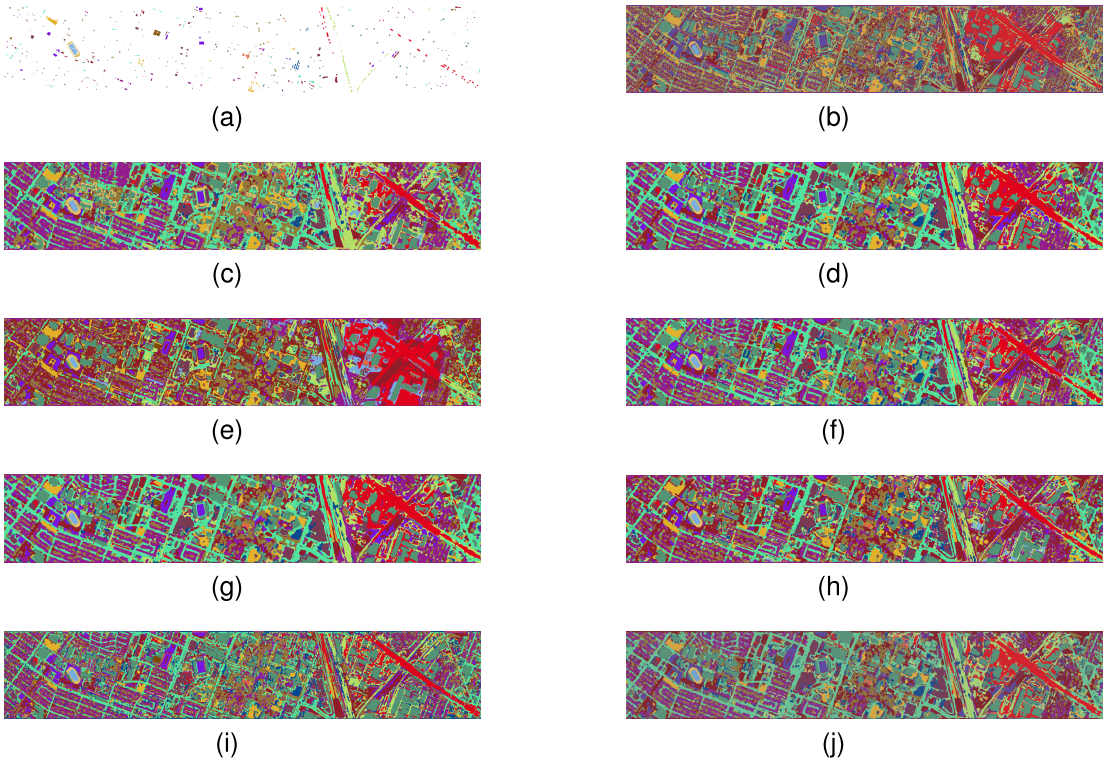


Fig. 6. Classification maps of different models on the HUST 2013 dataset. (a) Ground truth. (b) 3D-CNN. (c) SSAN. (d) MLNet-A. (e) ViT. (f) morphFormer. (g) MAEST. (h) MASSFormer. (i) SSFTT. (j) DFAN.

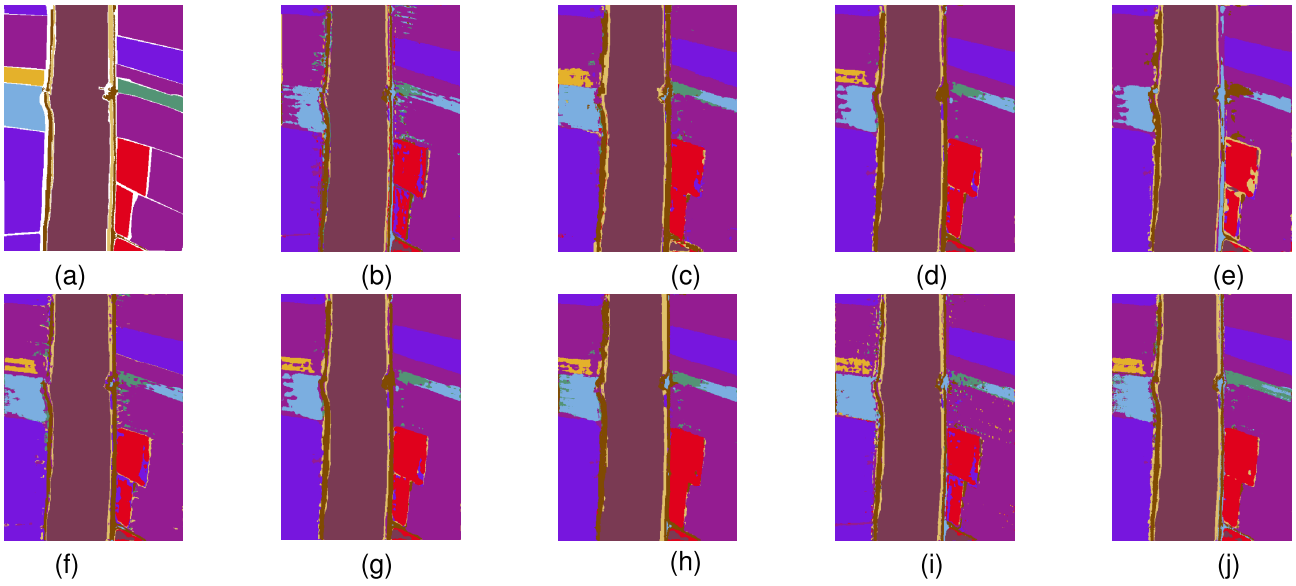


Fig. 7. Classification maps of different models on the WHU-Hi-Longkou dataset. (a) Ground truth. (b) 3D-CNN. (c) SSAN. (d) MLNet-A. (e) ViT. (f) morphFormer. (g) MAEST. (h) MASSFormer. (i) SSFTT. (j) DFAN.

time, it consistently outperforms other state-of-the-art HSI classification methods.

4) It can be found that ViT shows comparatively weaker classification performance than other methods on all HSI datasets. The main reason is that ViT fails to explore the local semantic features of HSIs. The CNN-based HSI classification models (i.e., 3D-CNN, SSAN, and LS^{2}CM) demonstrate superior performance on three datasets. This is because they take full advantage of the local spatial–spectral features of HSIs. These Transformer-based HSI classification models (i.e., ViT, morphFormer, LSGAViT, and MASSFormer) achieve superior performance on different HSI data. The main reason is that they effectively explore the local and global spatial–spectral features of HSIs by combining CNNs and Transformers. Nevertheless, our DFAN method obtains the highest classification performance among all methods. Compared with other methods, the

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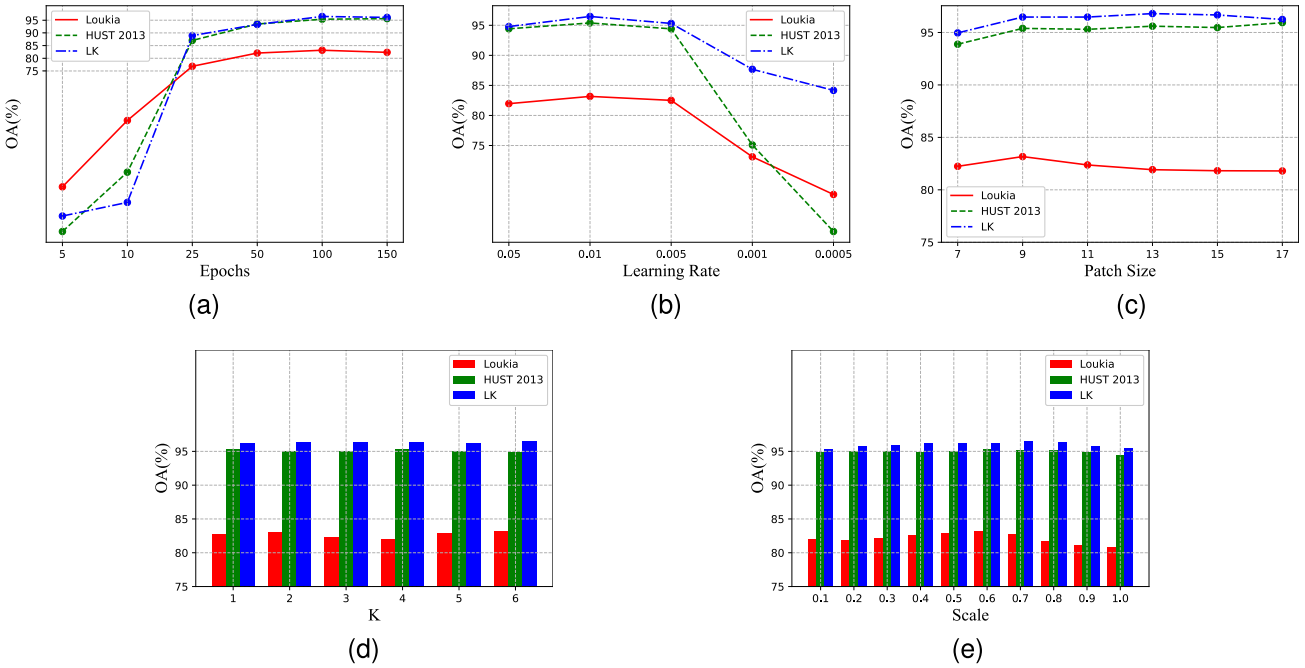


Fig. 8. OA values of our DFAN method with different parameters. (a) Training epochs. (b) Learning rate. (c) Patch size. (d) K in TLA-MLP. (e) Fusion scale.

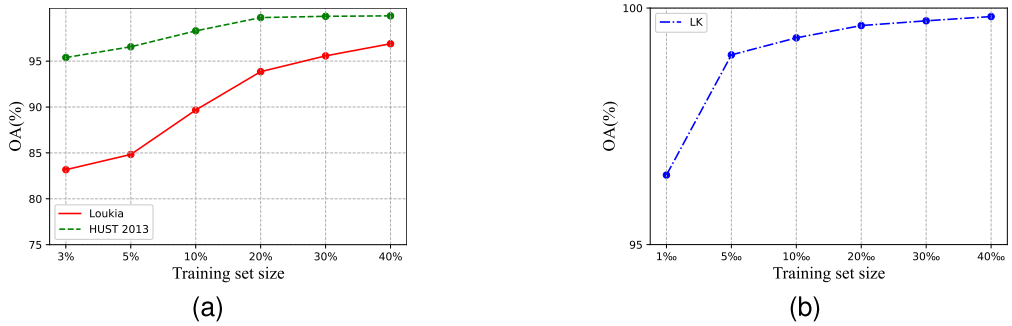


Fig. 9. OA values of the proposed DFAN method with different training sample sizes. (a) Loukia and HUST 2013 dataset. (b) WHU-Hi-Longkou datasets.

proposed DFAN method aggregates the spatial and spectral features of HSIs to further improve the classification performance. In addition, it performs multiscale neighboring token feature aggregation in the TLA module to improve the generalization capabilities of the model.

C. Parameter Analysis

To comprehensively evaluate the proposed DFAN model, we analyzed different hyperparameters involved in DFAN on classification performance, including training hyperparameters and structural hyperparameters. Fig. 8 shows the classification performances of our DFAN method with different hyperparameter settings on three HSI datasets. From these experimental results with different parameter settings on three datasets, we can get the following observations.

1) *Epochs:* We evaluate the performance of our proposed DFAN method using different training epochs from 5 to 150. Fig. 8(a) shows the classification performances of our DFAN method with different training epochs on three datasets. We can see that DFAN can only achieve OA of 29.34%, 11.74%, and 17.84% on Loukia, HUST 2013, and WHU-Hi-Longkou datasets with five training epochs, respectively. The proposed DFAN converges fast with the increase of training epochs and finally achieves the highest OA of 83.17%, 95.62%, and 96.47% at 100, 150, and 100 epochs on the Indian Pines, HUST 2013, WHU-Hi-Longkou datasets, respectively. As training time tends to increase with the growth of epochs, we set the training epochs for DFAN to 100 to balance training time and performance. To avoid unfair comparisons, we set the training epoch for the comparison methods to 100, considering the insufficient training of the comparison models.
2) *Learning Rate:* In this part, we investigate the influence of different learning rate settings in the proposed DFAN. The learning rate determines the convergence and convergence speed during training. Several commonly used learning rate settings are empirically tested and qualitatively compared on three datasets. The experimental results with different learning rate settings on three datasets are shown in Fig. 8(b). We can see that the

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TABLE IV
ABLATION EXPERIMENTS OF THE PROPOSED DFAN METHOD ON DIFFERENT DATASETS

Case	Component					HUST2013			LK			Loukia		
	Local Aggregation	TAP	Local	Global	Cross Atten	OA(%)	AA(%)	KAPPA(%)	OA(%)	AA(%)	KAPPA(%)	OA(%)	AA(%)	KAPPA(%)
1	×	✓	✓	✓	×	94.82±0.76	95.69±0.62	94.39±0.82	95.66±0.56	91.65±0.64	94.25±0.75	82.52±0.44	**84.98±1.26**	79.08±0.52
2	✓	×	✓	✓	×	95.08±0.33	95.65±0.30	94.68±0.36	96.14±0.48	92.09±1.24	94.90±0.63	83.12±0.52	84.13±3.61	79.81±0.60
3	✓	✓	✓	×	×	94.46±0.14	95.19±0.09	94.01±0.15	95.70±0.48	90.21±1.34	94.33±0.63	80.67±0.72	75.65±0.90	76.96±0.88
4	✓	✓	×	✓	×	93.05±0.64	94.33±0.53	92.49±0.70	82.84±5.14	42.17±19.44	74.79±7.69	79.98±0.93	76.83±1.90	76.02±1.15
5	×	×	✓	×	✓	93.24±0.65	94.52±0.42	92.69±0.70	95.43±0.63	90.53±1.39	93.95±0.84	82.60±0.92	83.29±1.60	79.16±1.10
6	✓	✓	✓	✓	×	**95.39±0.32**	**95.97±0.29**	**95.02±0.35**	**96.47±0.57**	**93.36±1.07**	**95.33±0.76**	**83.17±0.53**	83.94±1.63	**79.89±0.61**

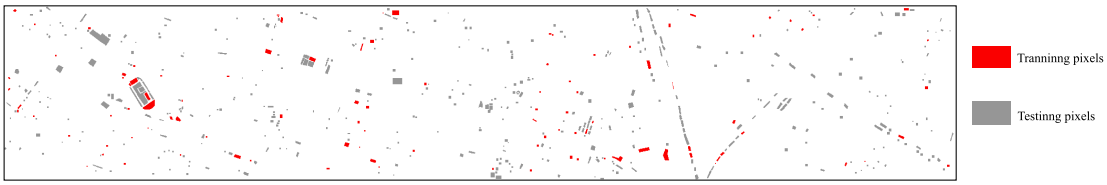


Fig. 10. Results of sample collection using FSS on the HUST 2013 dataset.

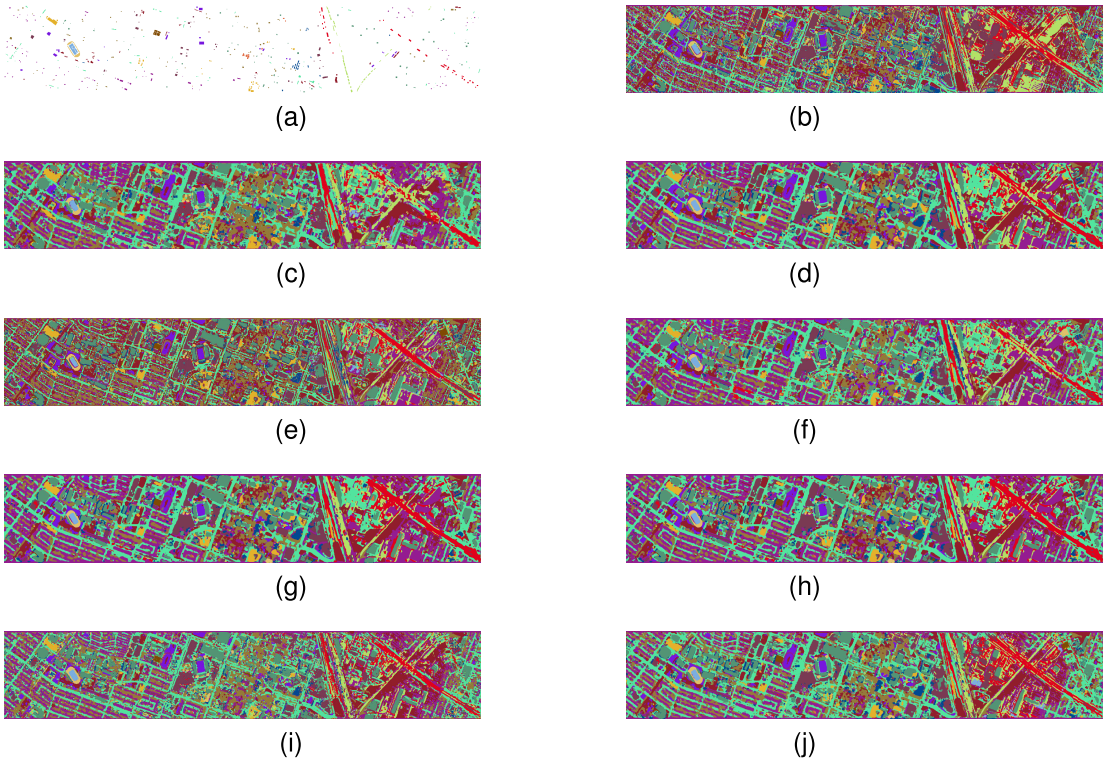


Fig. 11. Classification maps of different models on the HUST 2013 dataset with FSS. (a) Ground truth. (b) 3D-CNN. (c) SSAN. (d) MLNet-A. (e) ViT. (f) morphFormer. (g) MAEST. (h) MASSFormer. (i) SSFTT. (j) DFAN.

best classification accuracy is obtained on the Indian Pines (83.17%), HUST 2013 (95.39%), and WHU-Hi-Longkou (96.47%) datasets by setting the learning rate to 0.01. Therefore, we report all experimental results by setting the learning rating to 0.01.

3) *Patch Size:* As shown in Fig. 8(c), we investigate the effect of different spatial sizes of HSI samples on three datasets. It can be seen that the proposed model achieves the best OA with a patch size of 9 on the Loukia datasets. For the HUST 2013 and WHU-Hi-Longkou datasets, the model reaches the best OA with a patch size of 17 and 13. This variation is attributed to the diverse spatial–spectral complexity of the HSI dataset. Consequently, our experiments are conducted with a patch size of 9 to standardize the experimental settings.

4) *Number of Pooling Branches of TLA-MLP:* To explore the adaptability of TLA on different datasets, we verify the effect of the number of pooling branches in TLA on the classification performance. The range of K in TLA is set from 1 to 6. The results of the model with different numbers of pooling branches are shown in Fig. 8(d). It can be seen that the highest OA of 83.17%, 95.39%, and 96.47% is achieved on the Loukia, HUST 2013, and WHU-Hi-Longkou datasets when the selected values of K are 6, 1, and 6, respectively. We follow the above setting of K in subsequent experiments.

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TABLE V
SAMPLE SIZE OF THE HUST 2013 DATASET WITH FSS

No.	Class	Trainning Set	Testing Set
1	Healthy grass	146	1105
2	Stressed grass	460	794
3	Synthetic grass	91	606
4	Tree	186	1058
5	Soil	83	1159
6	Water	6	319
7	Residential	334	934
8	Commercial	160	1084
9	Road	306	946
10	Highway	177	1050
11	Railway	185	1050
12	Parking lot1	552	681
13	Parking lot2	44	425
14	Tennis court	79	349
15	Running track	385	275
Total		3194	11835

5) *Fusion Scale:* Fig. 8(e) shows the results of DFAN with different fusion scales set from 0.1 to 1. As the fusion scale increases, the percentage of global features in feature fusion rises gradually. The results show that the highest OA of 83.17%, 95.39%, and 96.47% is achieved on the Loukia, HUST 2013, and WHU-Hi-Longkou datasets when the value of the fusion scale is 0.6, 0.6, and 0.7, respectively.

D. Influence of Training Set Size

To validate the generalization performance of the proposed DFAN method, we carry out experiments to investigate how different training sample sizes affect classification accuracy. The classification results of the proposed model with different scaled training sets are shown in Fig. 9. Several training sets are randomly chosen from the Indian Pines, HUST 2013, and WHU-Hi-Longkou datasets. It can be seen that the performance of the proposed DFAN steadily increases as the number of training samples grows in the initial stage. For all three datasets, the incremental performance of the model decreases with an increase in the size of the training set when the size of the training set reaches a certain threshold. This is because the proposed DFAN method achieves relatively stable performance on all three datasets once it is trained with a sufficiently large number of training samples. This phenomenon further demonstrates the feasibility and stability of the DFAN method.

E. Ablation Study

In this section, we conduct ablation experiments on Indian Pines, HUST 2013, and WHU-Hi-Longkou datasets to demonstrate the effectiveness of the components in the proposed DFAN method. Here, four variants of the proposed DFAN method are constructed to investigate the effect of each component on the classification performance of the model. Specifically, for case 1, we discard the local aggregation (LA) module of the DFA module. For case 2, we remove the TLA module in the DFA module to verify its effectiveness. For case 3, the global branch in the DFA module is eliminated, and this variant only employs local spatial–spectral features to classify HSI pixels. For case 4, the situation is exactly the opposite, where the local branch in the DFA module is eliminated, and this variant only employed global spatial–spectral features to classify HSI pixels. For case 5, we replace the proposed global block with the cross-attention module. In the last case, all components of the HSI classification task are used in the original DFAN method.

Table IV illustrates the experimental results on three datasets, where bold indicates the highest value in each column. Compared with the original DFAN method, its variant without the LA mechanism exhibits a decrease in OA by 0.57%, 0.81%, and 0.65% on the HUST 2013, WHU-Hi-Longkou, and Loukia datasets, respectively. It can be seen that DFAN without LA is reduced in robustness. In addition, DFAN without the TLA module decreases the OA by 0.31%, 0.33%, and 0.05% on three datasets, respectively. It is also clear that the OA of DFAN without the global branch decreases by 0.93%, 0.77%, and 2.50% on three datasets, respectively. Similarly, the OA of DFAN without the local branch decreases by 2.34%, 13.63%, and 3.19% on three datasets, respectively. Finally, replacing the global block of DFAN with the cross-attention module results in a decrease in OA by 2.15%, 1.04%, and 0.57% on three datasets. This demonstrates the effectiveness of the global block in our proposed method. It can be seen that DFAN outperforms these variants on three HSI datasets across three metrics. We can easily conclude that removing any component from the original DFAN negatively affects the classification performance of our DFAN method. Therefore, the ablation experiments on three datasets can confirm the efficacy of the components in our proposed DFAN approach.

F. Effectiveness Evaluation With FSS

In the aforementioned experiments, the training samples and testing samples are randomly selected from the same HSIs, which is a common practice in HSI classification experiments. However, this approach makes it challenging to discern whether the improvement in classification accuracy results from the model's better use of spatial information or increased overlap between training and testing samples, potentially leading to biased performance evaluation. In this experiment, to minimize the overlap between samples, we adopt a fixed sampling strategy (FSS) [54] to evaluate the superiority of the proposed method. Fig. 10 shows the results of our sample collection on the HUST 2013 dataset. Table V shows the sample sizes of training and testing sets for HUST 2013 dataset using FSS.

To validate the superiority of the proposed model under the FSS condition, we adopt the same comparative models as previous experiments. Each HSI classification model is tested using the same training and testing datasets. Table VI shows the classification results of our DFAN model and its competitors on the HUST 2013 dataset with FSS. Fig. 11 illustrates the classification maps produced by various models on the HUST 2013 dataset. From these experimental results, it is easy to see that DFAN still achieves the highest classification performance among all HSI classification models. Specifically, our proposed DFAN method achieves the highest average OA value (81.32%), AA value (80.24%), and Kappa (79.77%).

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TABLE VI
RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS WITH FSS ON THE HUST 2013 DATASET

No.	DFAN	3D-CNN	SSAN	MLNet-A	ViT	morphFormer	MAEST	MASSFormer	SSFTT
1	99.24±0.51	99.44±0.43	99.13±0.88	99.43±0.41	99.71±0.13	99.69±0.38	99.41±0.52	98.80±0.82	98.85±1.79
2	71.08±2.80	68.15±1.72	67.96±3.37	74.77±0.27	74.42±6.47	72.51±1.45	75.10±0.44	69.48±3.45	62.11±3.95
3	99.02±0.99	92.98±6.75	98.95±0.65	100.00±0.00	86.91±1.10	93.55±4.27	100.00±0.00	97.10±5.41	97.81±1.71
4	99.14±0.54	95.82±4.41	94.83±1.11	97.78±0.55	93.10±4.80	97.94±0.68	97.96±0.56	95.32±1.85	98.16±0.68
5	95.38±1.51	97.91±1.64	95.53±1.34	93.99±1.50	95.05±0.93	96.72±1.42	95.61±1.02	94.84±1.01	96.30±2.45
6	40.00±54.77	0.00±0.00	0.00±0.00	28.56±41.47	21.76±37.12	0.00±0.00	38.22±42.97	29.00±41.70	15.22±17.25
7	81.64±6.71	79.86±4.95	68.74±4.92	69.82±2.84	63.50±3.34	63.91±3.57	69.54±1.73	64.36±3.77	70.73±2.31
8	59.65±1.94	72.06±9.79	59.79±8.86	59.28±3.27	56.42±1.73	60.72±6.46	56.66±1.91	73.89±6.09	75.63±9.98
9	82.39±4.91	69.93±4.45	74.30±5.28	69.68±3.47	78.78±2.92	80.33±4.18	68.47±6.35	70.51±3.60	77.28±4.18
10	76.31±8.83	74.48±4.63	71.90±7.18	72.92±3.19	81.91±2.65	77.15±3.47	75.27±5.24	78.54±6.25	81.64±3.94
11	71.23±11.01	64.01±11.76	72.10±3.48	84.41±3.63	81.29±6.91	63.54±8.89	92.80±4.04	64.04±4.95	67.05±7.58
12	58.57±2.09	43.77±5.01	58.85±5.86	62.20±2.33	48.00±3.66	55.50±4.26	60.71±1.88	53.71±5.29	49.62±4.48
13	85.81±4.18	76.93±3.36	67.12±8.11	85.96±2.48	82.17±5.97	79.26±11.82	86.34±2.83	90.47±2.29	90.40±5.53
14	93.46±7.28	97.24±1.91	74.96±11.29	94.90±3.96	73.44±1.25	83.72±13.23	92.13±3.74	81.55±2.90	95.10±2.69
15	90.72±1.94	92.56±4.09	93.17±3.33	95.30±0.34	94.03±1.06	74.70±16.09	93.72±3.94	92.06±2.01	91.09±4.32
OA(%)	**81.32±1.11**	76.34±0.92	76.65±1.78	80.58±0.51	77.17±0.41	77.30±2.06	80.68±0.81	77.50±2.04	78.42±1.43
AA(%)	**80.24±3.88**	75.01±0.60	73.16±1.33	79.27±2.99	75.37±2.29	73.28±2.56	80.13±2.88	76.91±3.89	77.80±2.17
Kappa(%)	**79.77±1.21**	74.38±0.99	74.69±1.93	78.97±0.56	75.27±0.44	75.43±2.23	79.09±0.88	75.65±2.20	76.64±1.55
FLOPs(G)	6.30	0.24	5.07	14.08	66.99	2.25	14.08	8.94	2.33
Params(M)	0.60	0.16	1.75	1.36	6.39	0.15	1.36	0.83	0.67
TrainTime(s)	336.33	75.37	314.33	179.15	151.68	255.18	147.98	147.08	96.90
TestTime(s)	4.29	1.26	3.15	1.63	1.11	3.49	1.06	1.10	1.20

The average OA value of the proposed DFAN model improves by 4.98% (3D-CNN), 4.67% (SSAN), 0.74% (MLNet-A), 4.15% (ViT), 4.02% (morphFormer), 0.64% (MAEST), 3.82% (MASSFormer), and 2.90% (SSFTT), respectively. Furthermore, our DFAN exhibits the fewest misclassification points and the smoothest boundaries between different categories in the feature map. This is because the proposed DFAN method aggregates the spatial and spectral features of HSIs to further improve the classification performance. In addition, it performs multiscale neighboring token feature aggregation in the TLA module to improve the generalization capabilities of the model. The results further demonstrate the superiority of our DFAN method in HSI classification tasks.

V. CONCLUSION

In this work, we propose a novel HSI classification framework, called DFAN. Unlike existing classification methods, our proposed model uses the interactive aggregation of global and local spatial–spectral information to extract intrinsic features from HSIs. To model the long-distance dependencies of HSIs, the TLA-MLP module is used to capture global position-channel-aware features. In addition, it allows each input global token to aggregate multiscale discriminative spatial–spectral semantic information from its local neighborhoods, thereby effectively improving the robustness of the model. To fully explore global and local spatial–spectral information, the local features extracted by the lightweight CNN are combined with the location-aware global feature weighting after self-global aggregation. Then, the cross-attention fusion of local and global spatial–spectral features can provide a more powerful classification capability for our DFAN method. The experimental results on three benchmark HSI datasets demonstrate that the proposed DFAN method outperforms other state-of-the-art HSI classification methods.

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